📈 Advertising Sales Prediction using Regression Models

📌 Project Report

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1. Introduction

Advertising plays a major role in increasing product sales. Companies invest money in different advertising platforms such as TV, Radio, and Newspaper to attract customers and improve sales performance.

This project focuses on predicting product sales based on advertising expenditure using Machine Learning regression techniques. Different regression models are trained and compared to understand which model performs best for sales prediction.

The project also analyzes feature importance, multicollinearity, and model performance using various evaluation metrics.

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2. Objective

The main objectives of this project are:

- To analyze the relationship between advertising expenditure and sales
- To build regression models for sales prediction
- To compare Linear Regression, Lasso Regression, and Ridge Regression
- To evaluate model performance using statistical metrics
- To study feature importance and multicollinearity

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3. Dataset Description

Dataset Used:

`Advertising.csv`

Features:

- **TV** → Advertising budget spent on TV
- **Radio** → Advertising budget spent on Radio

- **Newspaper** → Advertising budget spent on Newspaper

Target Variable:

- **Sales** → Product sales value

The dataset contains advertising expenditure data and corresponding sales values.

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4. Libraries Used

The following Python libraries were used in this project:

- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Statsmodels

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5. Methodology

5.1 Data Loading

The dataset was loaded using the Pandas library.

5.2 Data Understanding

Basic analysis was performed including:

- Dataset shape
- Dataset information
- Summary statistics
- Checking missing values
- Removing unnecessary columns

5.3 Feature and Target Separation

The dataset was divided into:

- Independent variables (X)
- Dependent variable (y)

5.4 Train-Test Split

The dataset was split into training and testing sets using Scikit-learn.

- Training Data → 80%
- Testing Data → 20%

5.5 Multicollinearity Analysis

Variance Inflation Factor (VIF) was calculated to check multicollinearity among features.

VIF helps identify whether independent variables are highly correlated.

5.6 Model Training

Three regression models were trained:

Linear Regression

A basic regression model used to predict continuous values.

Lasso Regression

A regularization technique that reduces less important feature coefficients.

Ridge Regression

A regularization technique that stabilizes coefficients and reduces overfitting.

5.7 Prediction

Predictions were generated using all trained models.

5.8 Model Evaluation

Models were evaluated using:

- R^2 Score
- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)

5.9 Coefficient Analysis

Feature coefficients were compared across different regression models.

5.10 Visualization

Scatter plots were created to compare:

- Actual Sales
- Predicted Sales

5.11 Cross Validation

5-Fold Cross Validation was applied to evaluate model generalization and reliability.

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6. Regression Models Used

6.1 Linear Regression

Linear Regression establishes a linear relationship between advertising expenditure and sales.

Advantages:

- Simple and interpretable
- Fast training
- Good baseline model

6.2 Lasso Regression

Lasso Regression applies L1 regularization.

Advantages:

- Performs feature selection
- Reduces less important coefficients
- Helps prevent overfitting

6.3 Ridge Regression

Ridge Regression applies L2 regularization.

Advantages:

- Reduces coefficient variance
- Improves model stability
- Handles multicollinearity effectively

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7. Evaluation Metrics

R^2 Score

Measures how well the model explains variance in the target variable.

Higher R^2 score indicates better model performance.

Mean Absolute Error (MAE)

Measures average absolute prediction error.

Lower MAE indicates better accuracy.

Mean Squared Error (MSE)

Measures average squared prediction error.

Lower MSE indicates better model performance.

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8. Results and Analysis

- Linear Regression achieved strong prediction accuracy.
- Lasso Regression reduced the effect of less important features.
- Ridge Regression stabilized coefficients and reduced overfitting.
- TV advertising had the highest impact on sales prediction.
- Cross-validation results confirmed model consistency and reliability.

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9. Key Insights

- TV advertising contributes significantly to sales.
- Radio advertising also influences sales positively.
- Newspaper advertising showed comparatively lower impact.
- Regularization techniques improve model stability.
- Multicollinearity analysis helped understand feature relationships.

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10. Conclusion

This project successfully implemented multiple regression models for predicting product sales based on advertising expenditure.

Among the models used, Linear Regression performed very effectively, while Lasso and Ridge Regression improved coefficient handling and model stability through regularization techniques.

The project demonstrates how Machine Learning can help businesses make data-driven advertising decisions and improve marketing strategies.

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11. Future Enhancements

Future improvements for this project may include:

- Using larger real-world datasets
- Applying advanced regression models
- Hyperparameter tuning
- Feature engineering
- Deploying the model as a web application

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12. Project Structure

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OIBSIP_Advertising_Sales_Prediction/
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├── Advertising_Sales_Prediction.ipynb
├── Advertising.csv
├── README.md
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13. References

- Scikit-learn Documentation
- Pandas Documentation
- Statsmodels Documentation
- Machine Learning Regression Concepts

☑ Outcome

Successfully developed and evaluated multiple regression models capable of predicting sales using advertising expenditure data with good accuracy and interpretability.